DATA ANALYTICS PORTFOLIO

NEXT Retail

Customer Analytics & RFM Segmentation

Two years of orders turned into a clear answer: which customers to protect, which to win back, and which to welcome.

5,879 customers | 17.37M revenue | 50.84% churn

PostgreSQL | Power BI | RFM Analysis

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Data Analyst | SQL & Power BI



The dashboard home page

PROJECT AT A GLANCE

The Business Problem

Why this analysis was needed

The situation

NEXT is a UK online-only retailer of gifts and home decor, and many of its customers are wholesale buyers.

Leadership had no clear view of which customers drive revenue or which ones are quietly leaving. Marketing was reactive, and every customer got the same treatment.

Goal: find the most valuable customer segments, measure churn risk, and tell marketing exactly what to do with each group.

5,879
customers scored

17.37M
total revenue

779K
clean transaction rows

2 years
Dec 2009 to Dec 2011

Questions this project answers

1. Who are the most valuable customers?
2. Who has gone quiet and may churn?
3. How do value and churn change over time?
4. How do countries differ?
5. What should be done for each segment?

THE METHOD

How I Built It

From raw transactions to customer segments

1 Clean in SQL

Removed duplicates, missing customer IDs, returns and test products.

2 Score with RFM

Recency, Frequency and Monetary value for every customer, scored 1 to 5.

3 Segment and flag churn

Six segments. Churned means no purchase in 90 days.

4 Visualize in Power BI

A 3-page interactive dashboard with filters for time, country and segment.

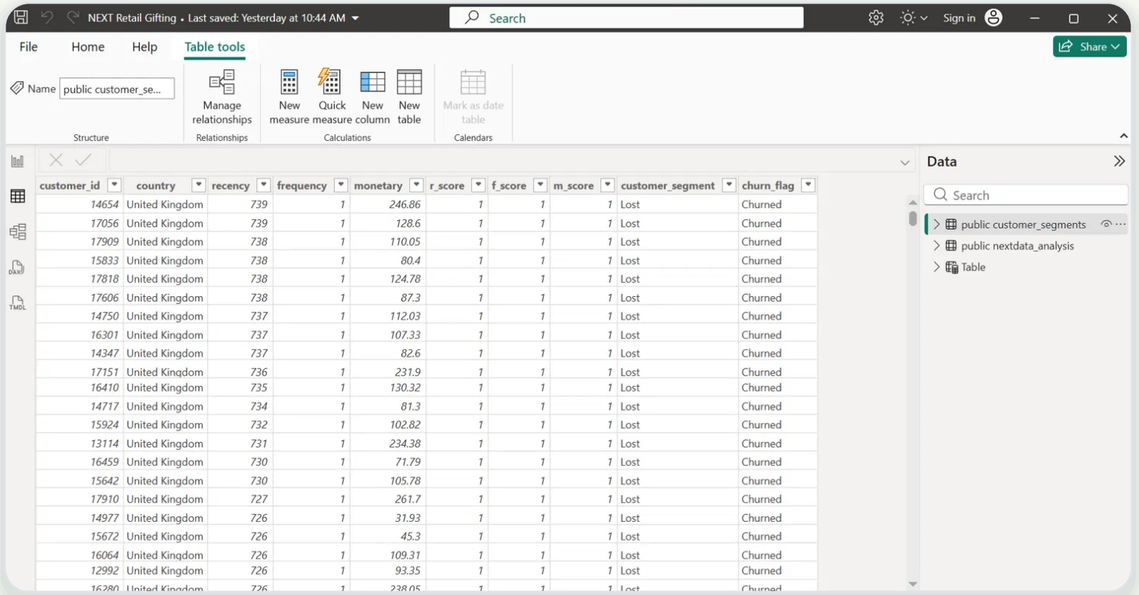
Cleaning at a glance

Raw rows — 1,067,371

Analysis rows — 779,483

- Exact duplicate rows removed
- Rows without a customer ID removed
- Returns and cancellations removed
- Test products removed
- Dates converted from text to timestamps

Customer-level output: one row per customer with R, F, M scores, segment and churn flag



customer_id	country	recency	frequency	monetary	r_score	f_score	m_score	customer_segment	churn_flag
14654	United Kingdom	739	1	246.86	1	1	1	Lost	Churned
17056	United Kingdom	739	1	128.6	1	1	1	Lost	Churned
17909	United Kingdom	738	1	110.05	1	1	1	Lost	Churned
15833	United Kingdom	738	1	80.4	1	1	1	Lost	Churned
17818	United Kingdom	738	1	124.78	1	1	1	Lost	Churned
17606	United Kingdom	738	1	87.3	1	1	1	Lost	Churned
14750	United Kingdom	737	1	112.03	1	1	1	Lost	Churned
16301	United Kingdom	737	1	107.33	1	1	1	Lost	Churned
14347	United Kingdom	737	1	82.6	1	1	1	Lost	Churned
17151	United Kingdom	736	1	231.9	1	1	1	Lost	Churned
16410	United Kingdom	735	1	130.32	1	1	1	Lost	Churned
14717	United Kingdom	734	1	81.3	1	1	1	Lost	Churned
15924	United Kingdom	732	1	102.82	1	1	1	Lost	Churned
13114	United Kingdom	731	1	234.38	1	1	1	Lost	Churned
16459	United Kingdom	730	1	71.79	1	1	1	Lost	Churned
15642	United Kingdom	730	1	105.78	1	1	1	Lost	Churned
17910	United Kingdom	727	1	261.7	1	1	1	Lost	Churned
14977	United Kingdom	726	1	31.93	1	1	1	Lost	Churned
15672	United Kingdom	726	1	45.3	1	1	1	Lost	Churned
16064	United Kingdom	726	1	109.31	1	1	1	Lost	Churned
12992	United Kingdom	726	1	93.35	1	1	1	Lost	Churned

THE METHOD

RFM Explained

Three questions that describe every customer

R — Recency

How recently did they buy?

MEASURED AS
Days since last order, counted from 10 Dec 2011

SCORED
Scored 1 to 5 in equal-sized groups

F — Frequency

How often do they buy?

MEASURED AS
Number of distinct invoices

SCORED
Fixed bands: 1, 2, 3-4, 5-9, 10+ orders

M — Monetary

How much do they spend?

MEASURED AS
Total revenue (quantity x price)

SCORED
Scored 1 to 5 in equal-sized groups

How customers become segments

Champions

R 4-5, F 4-5, M 4-5

Recent, frequent, big spenders

Loyal

R 3+, F 3+

Steady buyers

At Risk

R 1-2, F 3+

Used to buy often, now quiet

Lost

R 1-2, F 1-2

Inactive for a long time

Needs Attention

Everyone else

Recent but infrequent

New

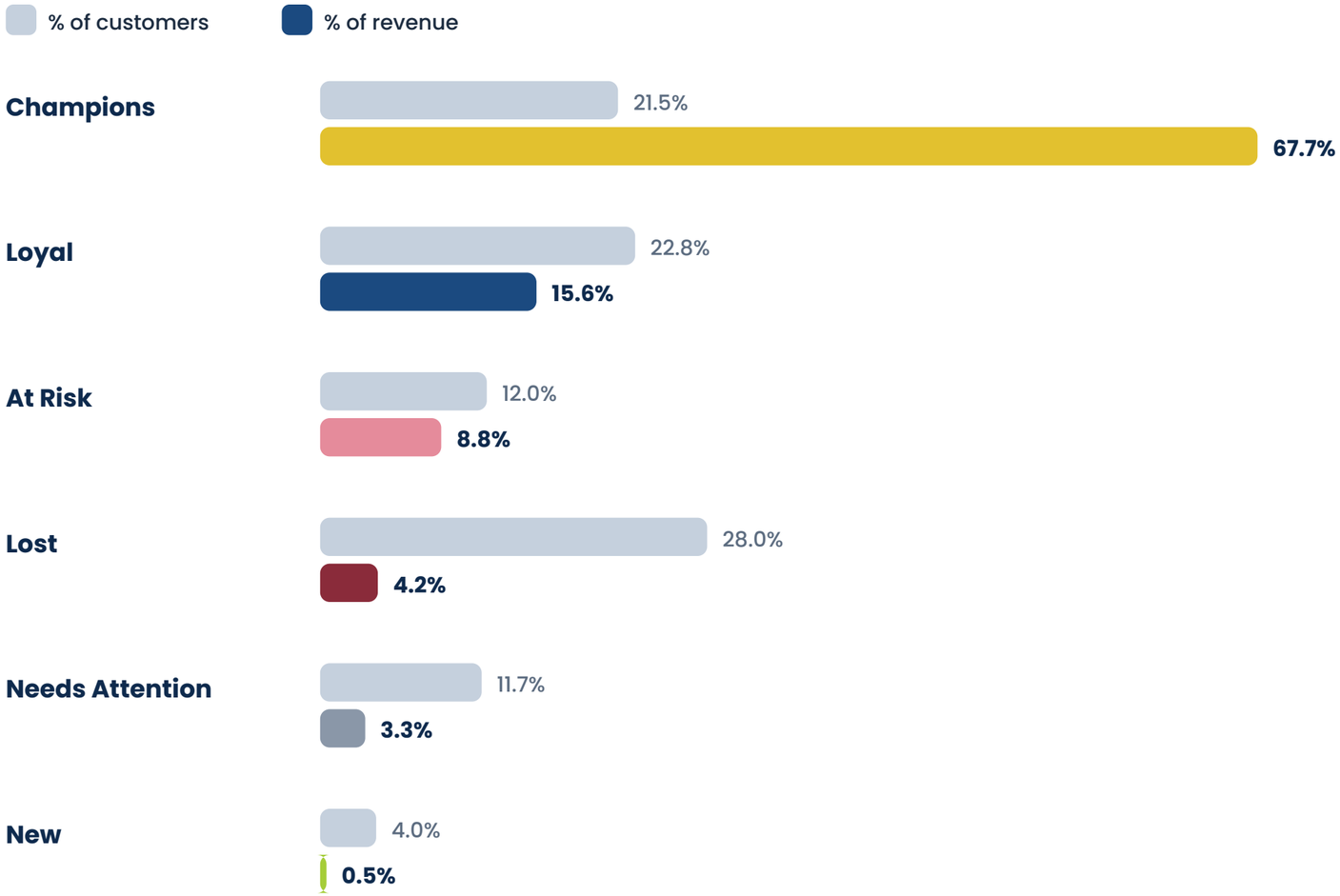
R 4-5, F = 1

Recent first-time buyers

FINDINGS

A Small Group Drives the Business

Share of customers vs share of revenue, by segment



21.5%

of customers (Champions)

67.7%

of total revenue

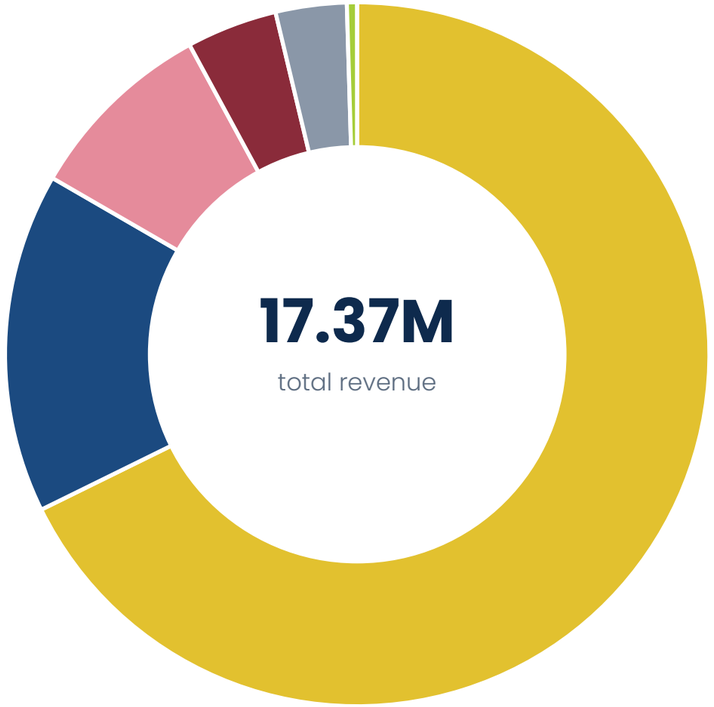
At Risk and Lost together make up 40% of customers but only about 13% of revenue.

Protecting Champions matters more than any other single action.

FINDINGS

Where the Revenue Comes From

17.37M in total revenue, split by segment

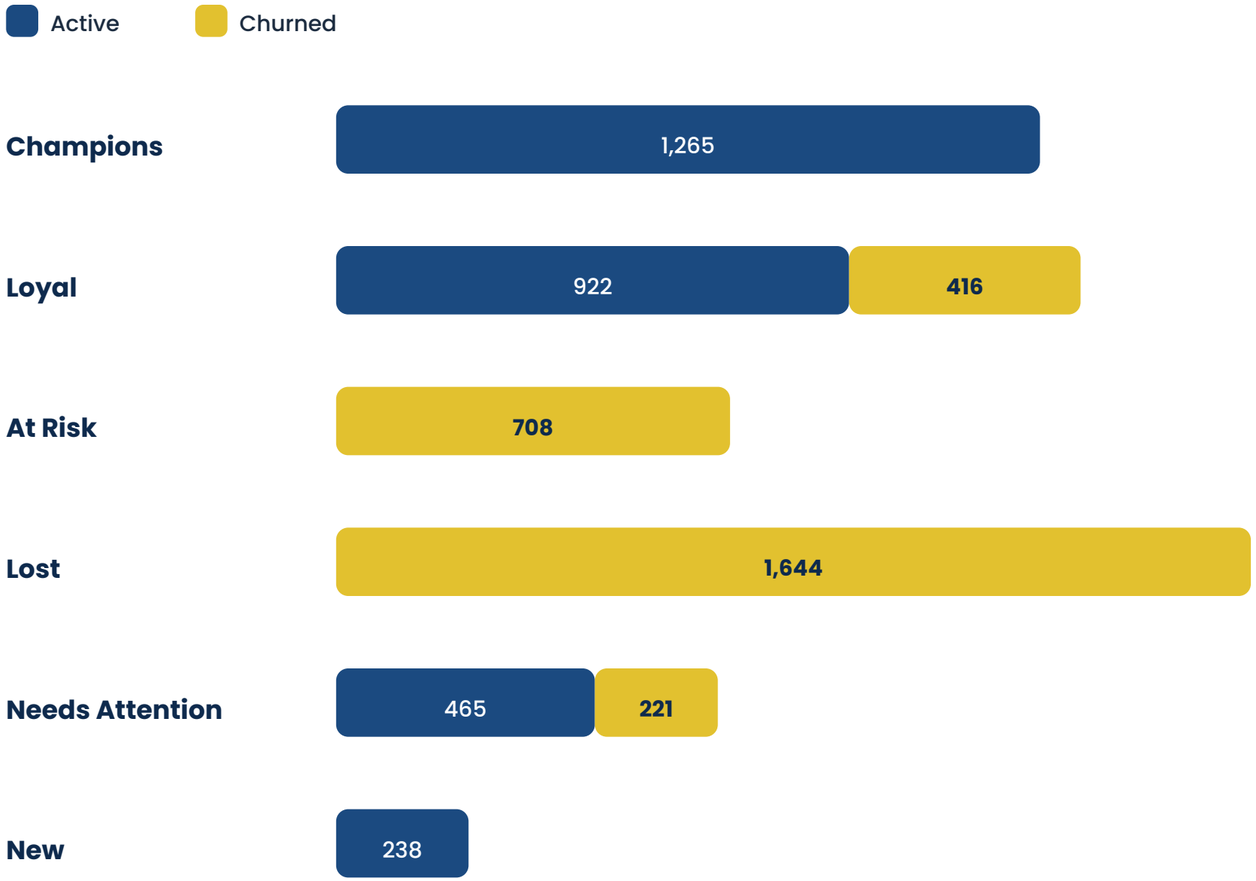


Segment	CUSTOMERS	REVENUE	SHARE
Champions	1,265	11.77M	**67.7%**
Loyal	1,338	2.72M	**15.6%**
At Risk	708	1.52M	**8.8%**
Lost	1,644	0.73M	**4.2%**
Needs Attention	686	0.57M	**3.3%**
New	238	0.08M	**0.5%**
Total	**5,879**	**17.37M**	**100%**

FINDINGS

Who Has Gone Quiet

Active vs churned customers (no purchase in 90 days)



50.84%

of customers have not bought in 90 days (about 2,990 people)

~640

churned customers sit inside Loyal and Needs Attention, so they are easy to miss

~2.25M

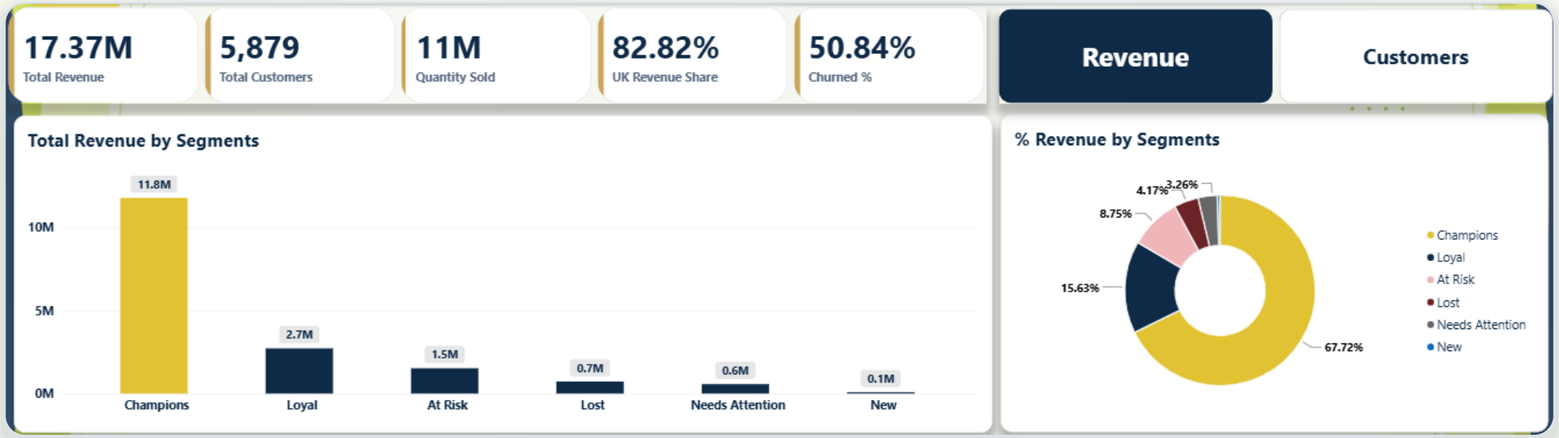
past revenue still tied to At Risk and Lost customers

THE DASHBOARD

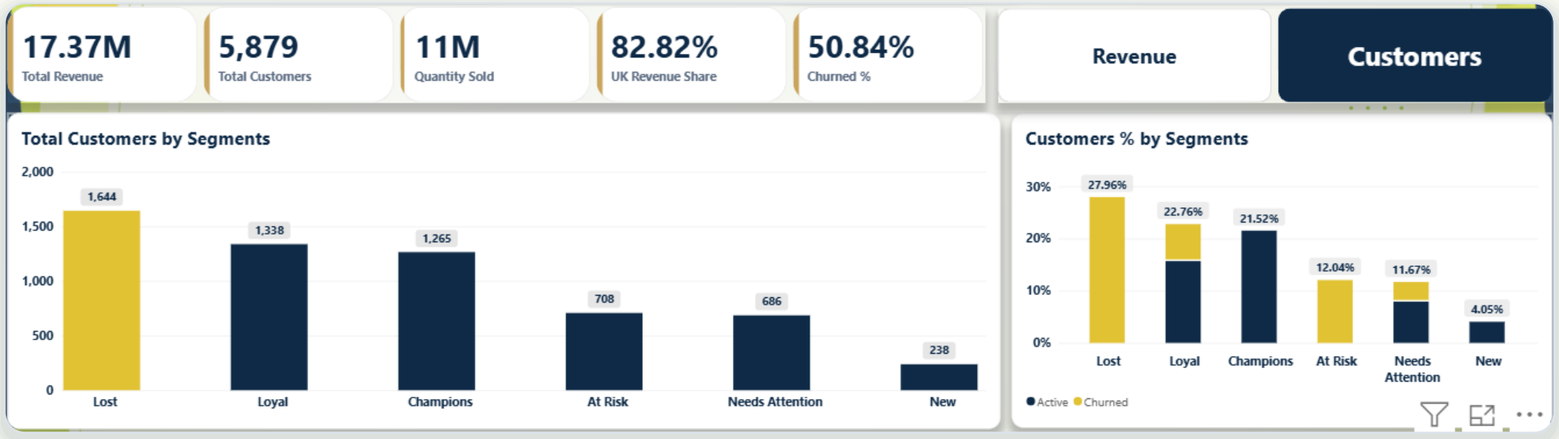
Revenue & Customers Overview

Switch between the Revenue and Customers views, and filter by month, quarter, year, country, segment and churn status

REVENUE VIEW



CUSTOMERS VIEW



Revenue view

- Total revenue, customers, quantity sold and UK share at a glance
- Revenue by segment as bars and a share chart

Customers view

- Customers per segment, split into active and churned
- Trends by month, quarter and year sit underneath

Both views react to every filter, so any slice of the business is one click away.

THE DASHBOARD

RFM & Country Analysis

Segment performance, market performance and churn trends

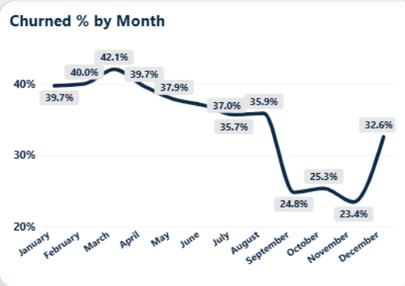
SEGMENT PERFORMANCE

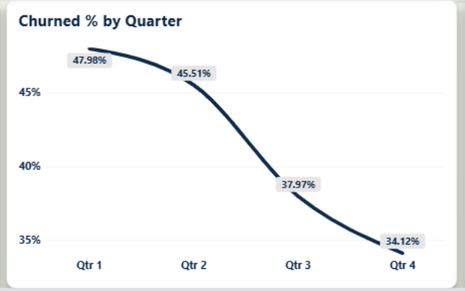
Segment	Total Customers	Total Revenue	Churned %	Avg.of R Score	Avg.of F Score	Avg.of M Score
Champions	1,265	11,765,377.61		4.58	4.61	4.65
Loyal	1,338	2,716,443.09	31.09%	3.62	3.57	3.47
At Risk	708	1,521,050.14	100.00%	1.69	3.48	3.39
Lost	1,644	725,082.42	100.00%	1.42	1.31	1.74
Needs Attention	686	566,119.55	32.22%	3.56	1.64	2.15
New	238	80,505.44		4.32	1.00	1.57

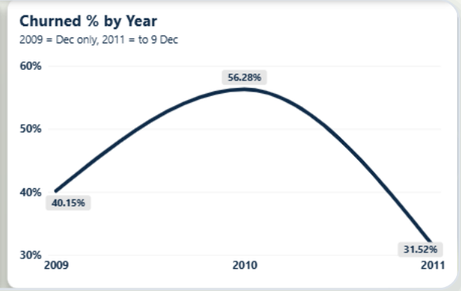
COUNTRY PERFORMANCE

Country	Total Customers	Total Revenue	Churned %	Avg.of R Score	Avg.of F Score	Avg.of M Score
United Kingdom	5,351	14,389,210.46	51.15%	2.99	2.83	2.96
EIRE	4	616,368.98	50.00%	3.50	4.00	4.00
Netherlands	22	554,038.09	72.73%	2.18	2.68	3.05
Germany	107	425,939.69	35.51%	3.42	3.02	3.61
France	93	347,090.88	35.48%	3.54	2.84	3.20
Australia	13	158,775.57	61.54%	2.38	2.54	2.77
Spain	41	120,361.63	43.90%	3.07	2.46	2.95

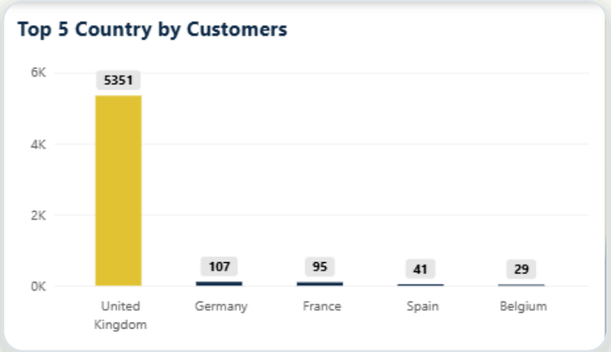
CHURN TRENDS







TOP COUNTRIES BY CUSTOMERS



Both tables are one toggle apart: Segment or Country.

FINDINGS

Markets Worth Watching

The UK is 82.8% of revenue, so the other markets are shown at their own scale

Revenue by country, excluding the UK (thousands)

COUNTRY		CUSTOMERS	CHURNED
EIRE	616K	4	50.0%
Netherlands	554K	22	72.7%
Germany	426K	107	35.5%
France	347K	93	35.5%
Australia	159K	13	61.5%
Spain	120K	41	43.9%

United Kingdom: 5,351 customers, 14.39M revenue, 51.15% churned

EIRE

4 customers, 616K revenue

About 154K each. A handful of very valuable accounts worth protecting.

Netherlands

22 customers, 72.7% churn

Over half a million in revenue, but most of these customers have gone quiet.

United Kingdom

5,351 customers, 82.8% of revenue

The core market. Retention here moves the whole business.

RECOMMENDATIONS

Actions & Business Decisions

A different plan for each customer segment

Champions — PROTECT

1,265 customers | 67.7% of revenue

ACTION

VIP rewards, early access and thank-you gifts.

DECISION

Protect first. Give the biggest accounts a dedicated contact.

Loyal — GROW

1,338 customers | 15.6% of revenue

ACTION

Bundles and repeat-order incentives.

DECISION

Build a path to Champion. Re-engage the ~416 who went quiet.

At Risk — WIN BACK

708 customers | 8.8% of revenue

ACTION

Personal win-back offers and discounts.

DECISION

Act fast. Best win-back return before they become Lost.

Lost — TRY ONCE

1,644 customers | 4.2% of revenue

ACTION

One low-cost "we miss you" campaign.

DECISION

Cap the spend. Stop if there is no response.

Needs Attention — CONVERT

686 customers | 3.3% of revenue

ACTION

Targeted offers and product suggestions.

DECISION

Turn occasional buyers into second and third orders.

New — WELCOME

238 customers | 0.5% of revenue

ACTION

Welcome message and a small gift.

DECISION

Win the second order, where repeat customers are made.

WRAPPING UP

Limits, Next Steps and Contact

What to keep in mind

- Churn and segments both use recency, so At Risk and Lost are 100% churned by design.
- Churn trends fall toward 2011 because recent buyers have not had time to go quiet.
- Wholesale and retail customers are not separated in the data.

What I would do next

- Separate wholesale from retail customers
- Add a revenue-at-risk view
- Add customer lifetime value
- Build a predictive churn model

Want insights like this from your customer data?

I turn raw transactions into segments, dashboards and clear next steps.

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Full project, SQL and dashboard on GitHub